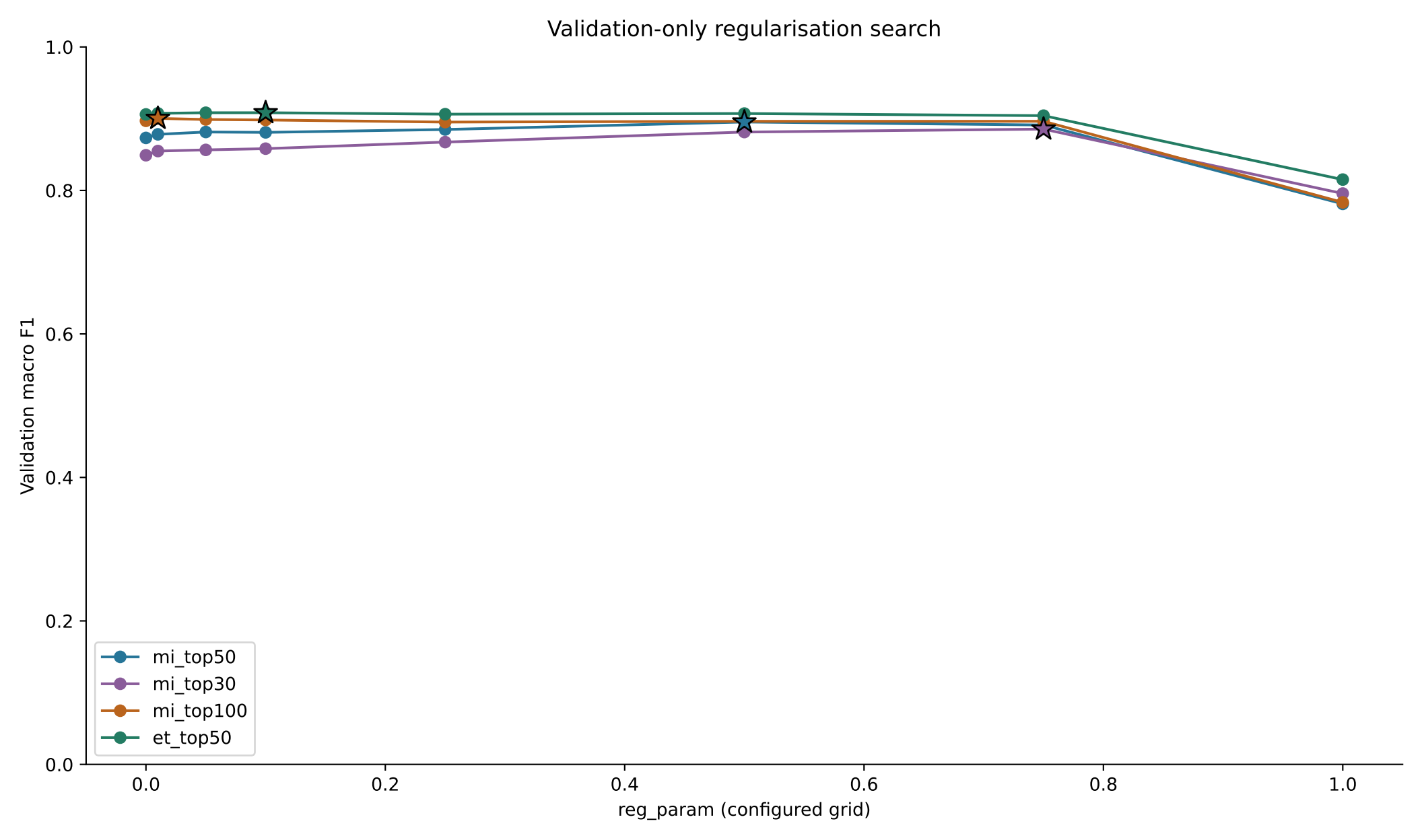
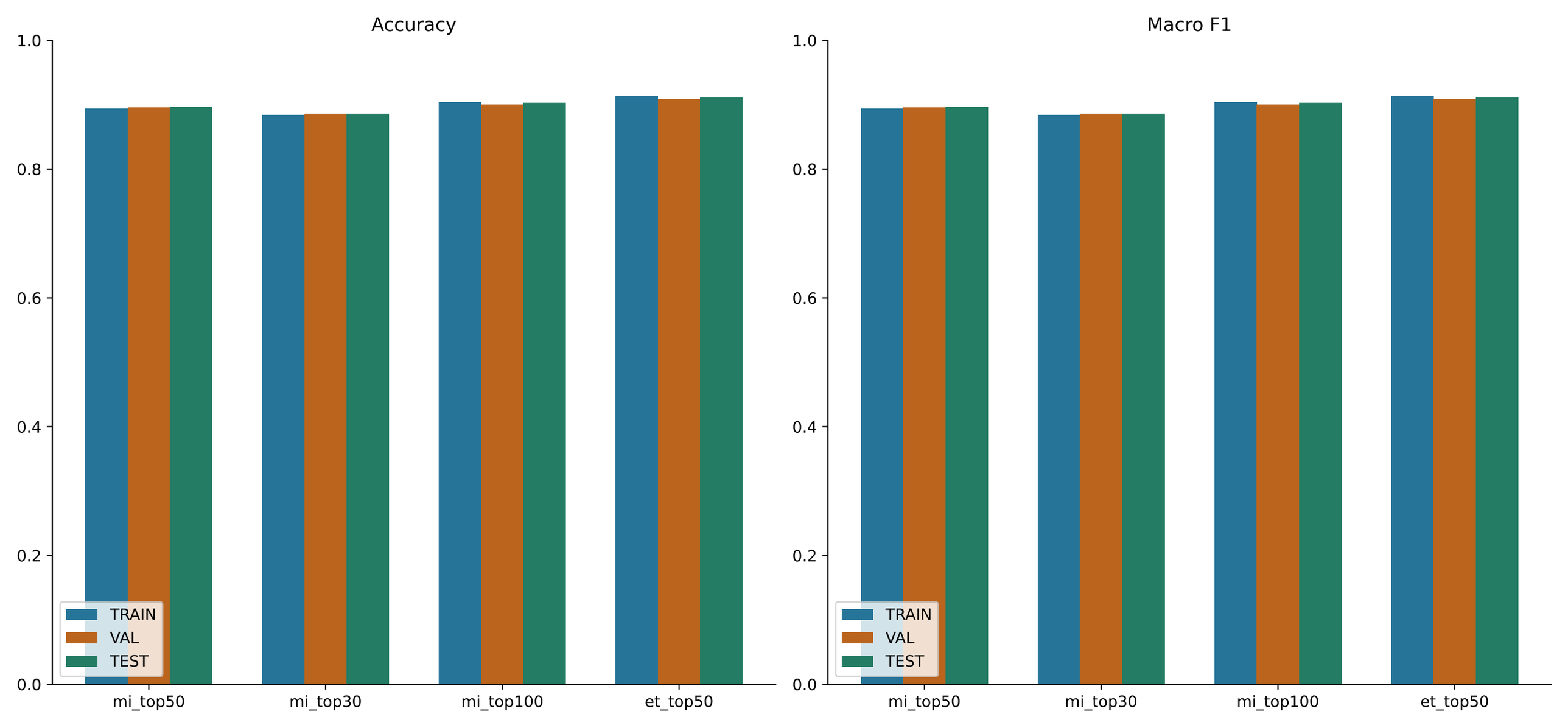
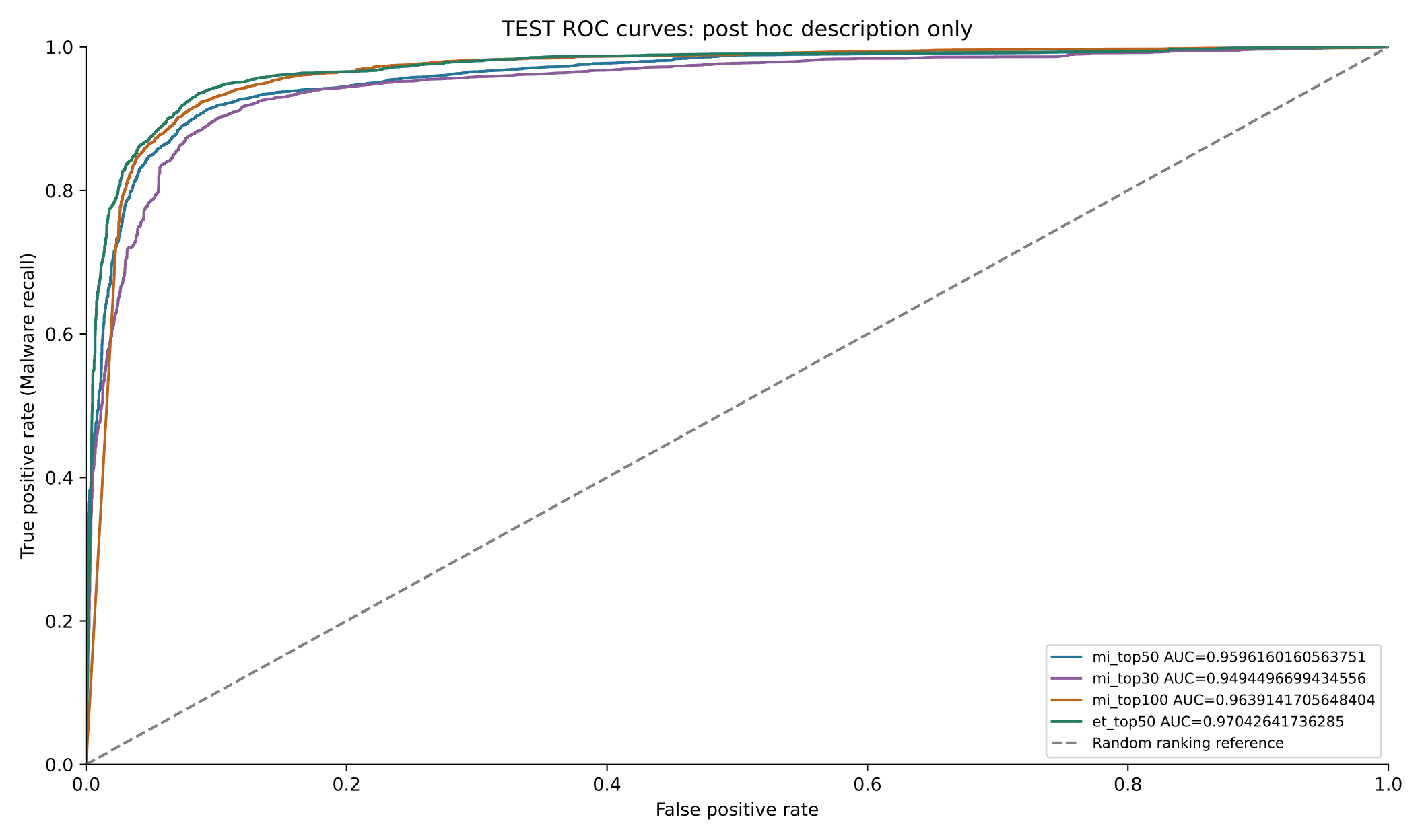
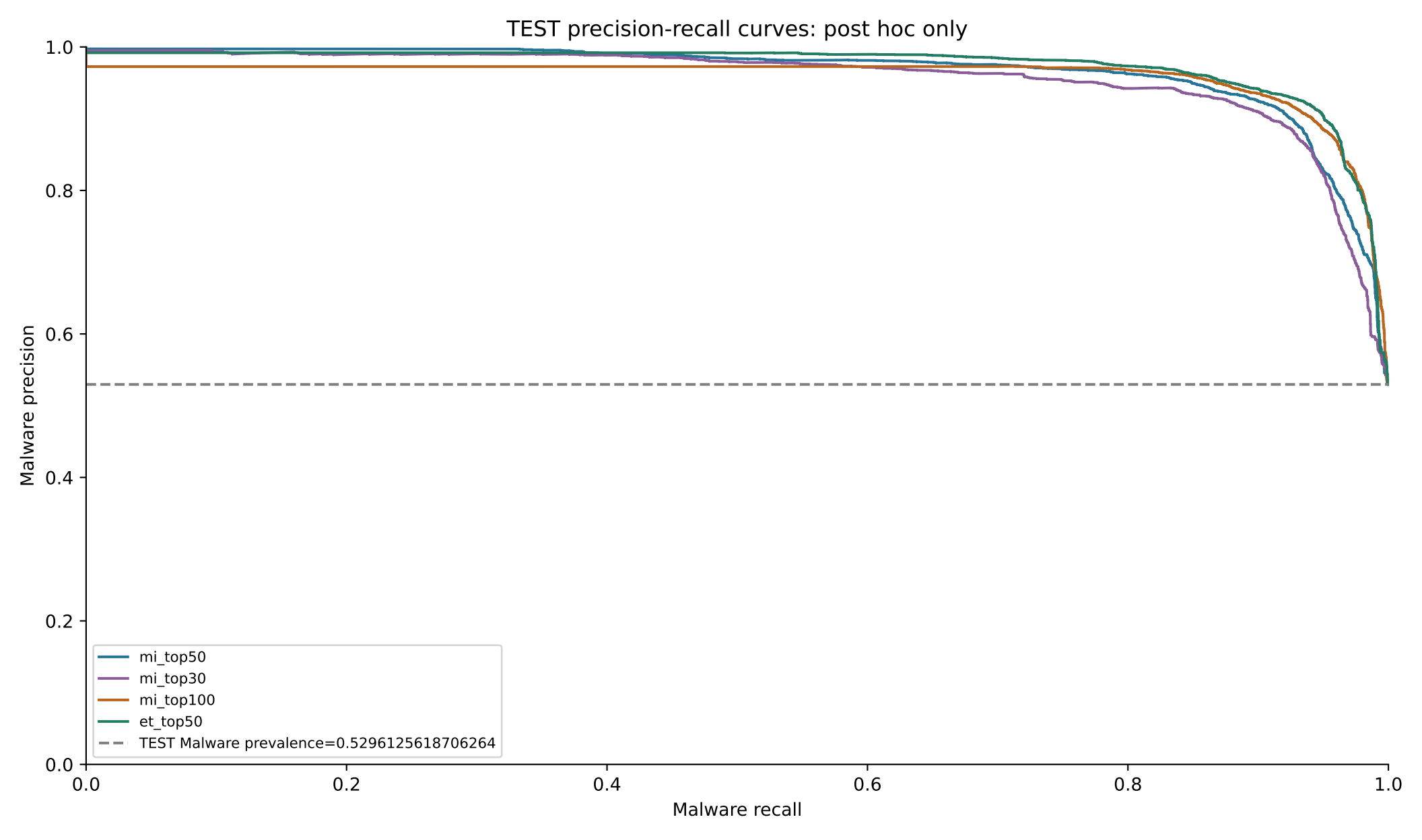
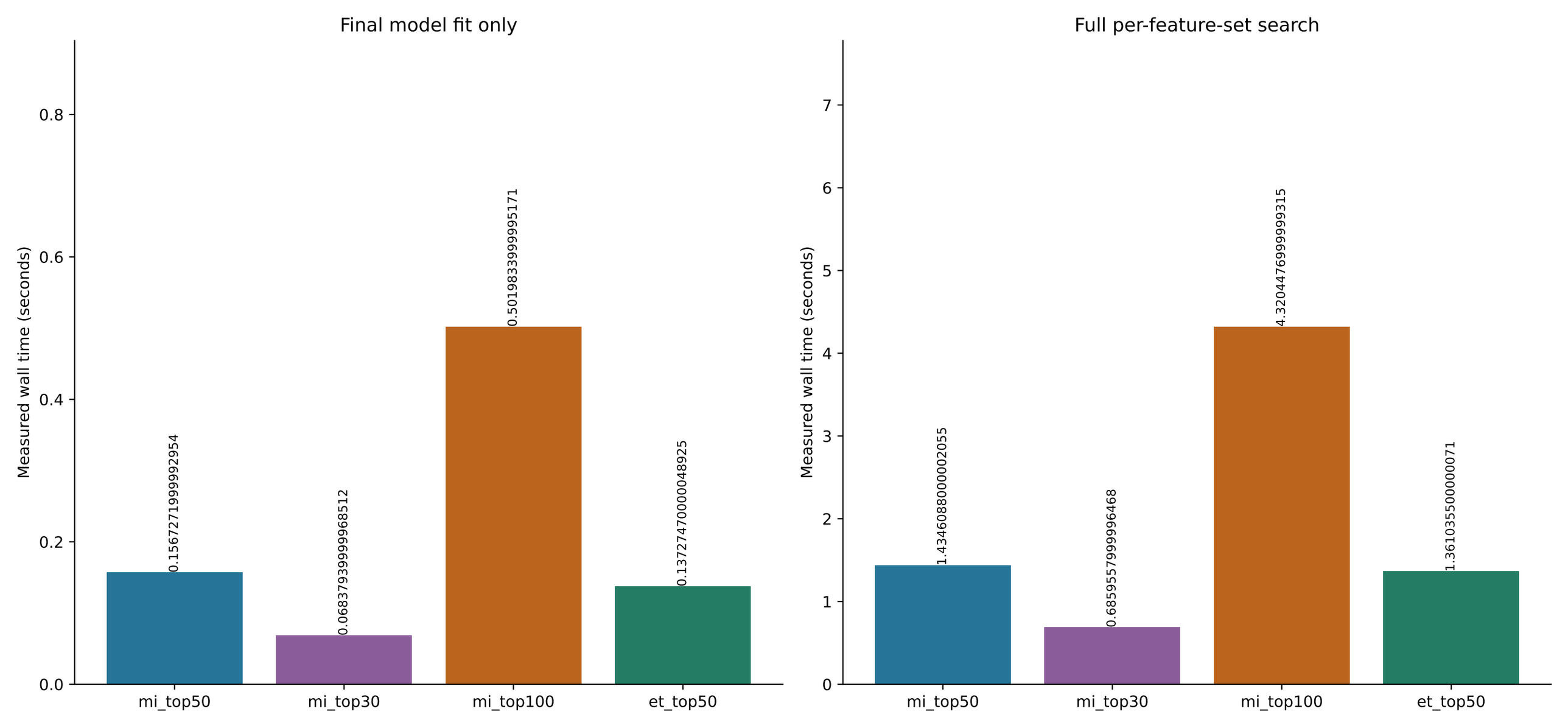
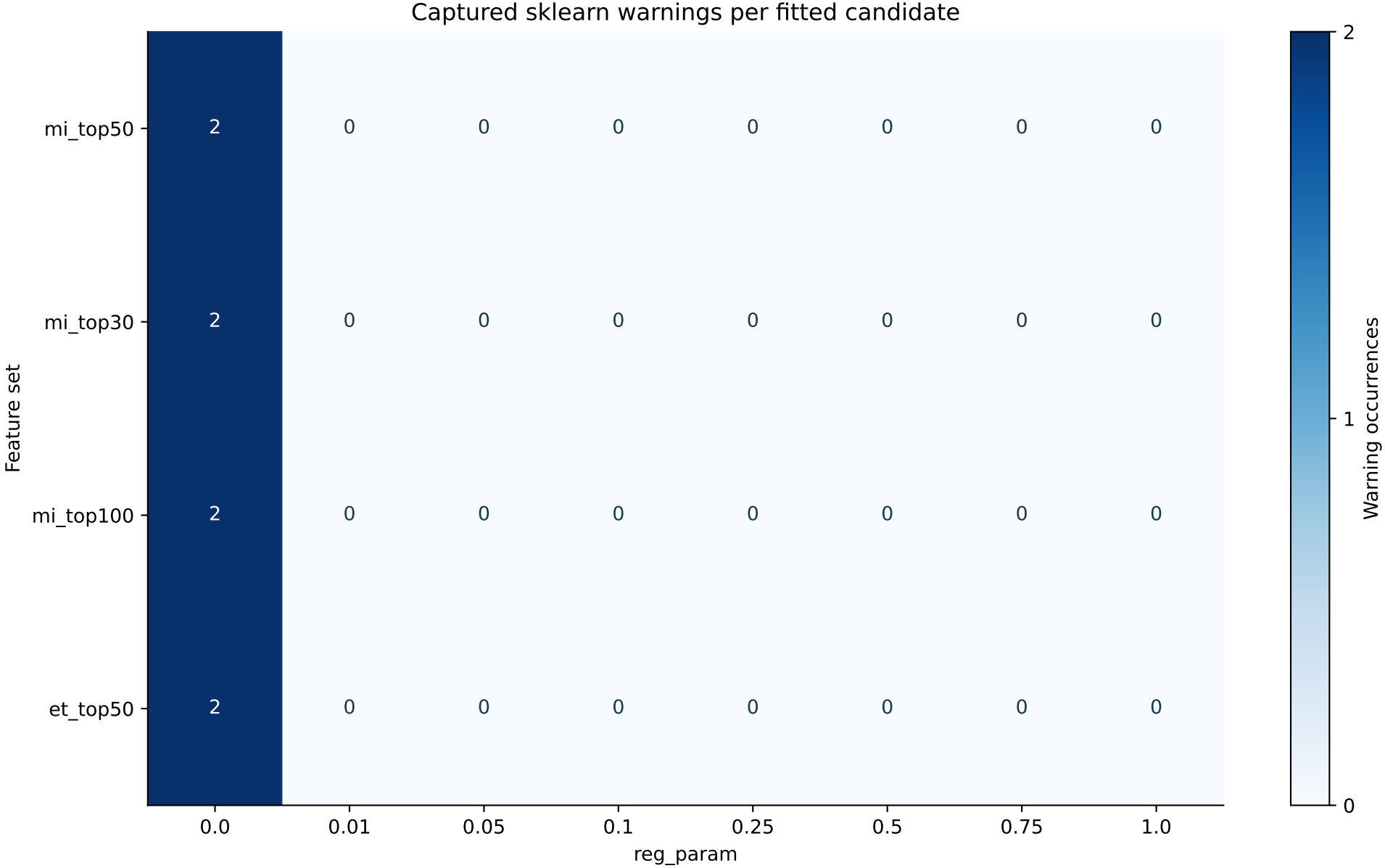

Validation-only regularisation search
Validation macro F1
reg_param (configured grid)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
mi_top50
mi_top30
mi_top100
et_top50

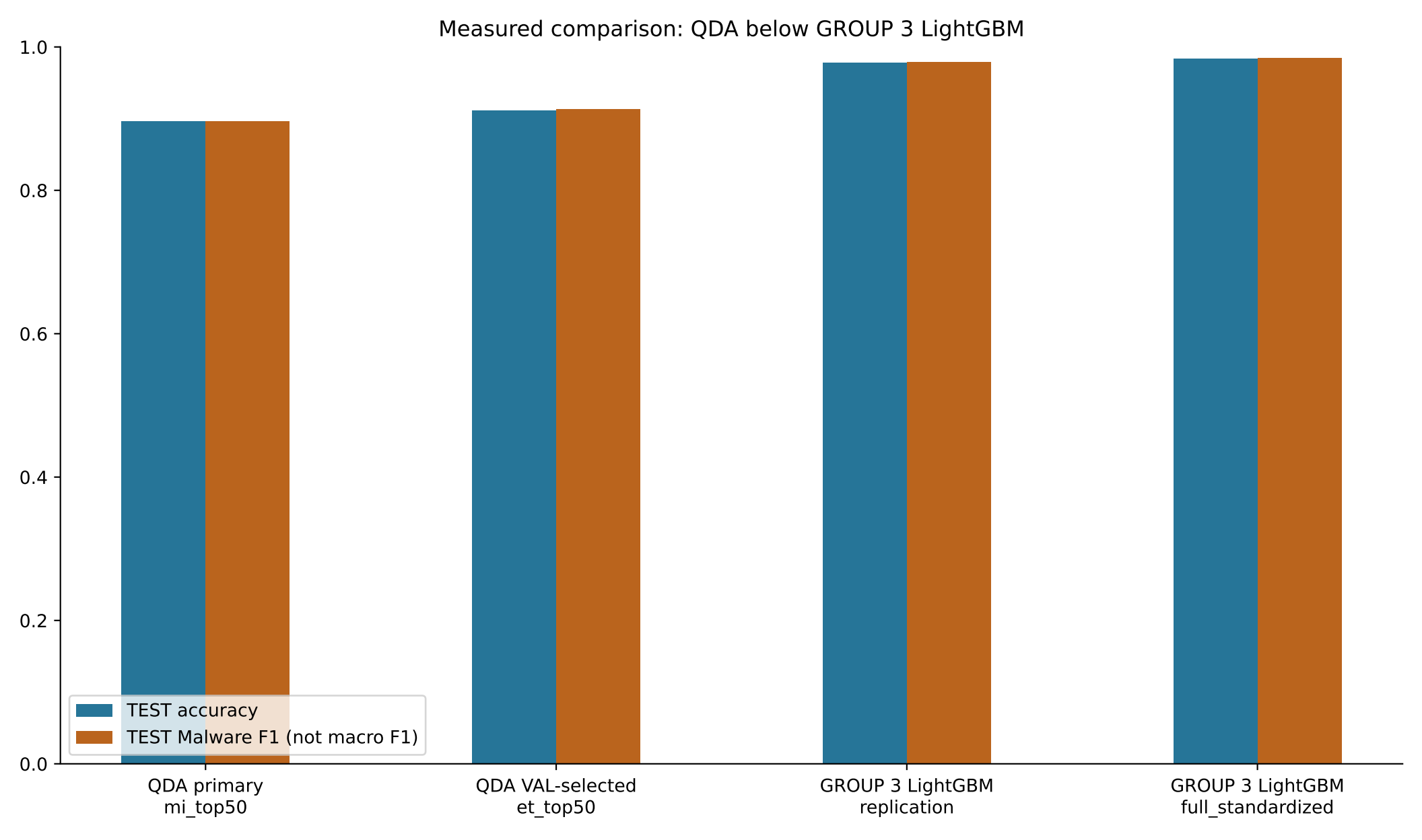

Accuracy
Macro F1
1.0
0.8
0.6
0.4
0.2
0.0
TRAIN
VAL
TEST
mi_top50
mi_top30
mi_top100
et_top50

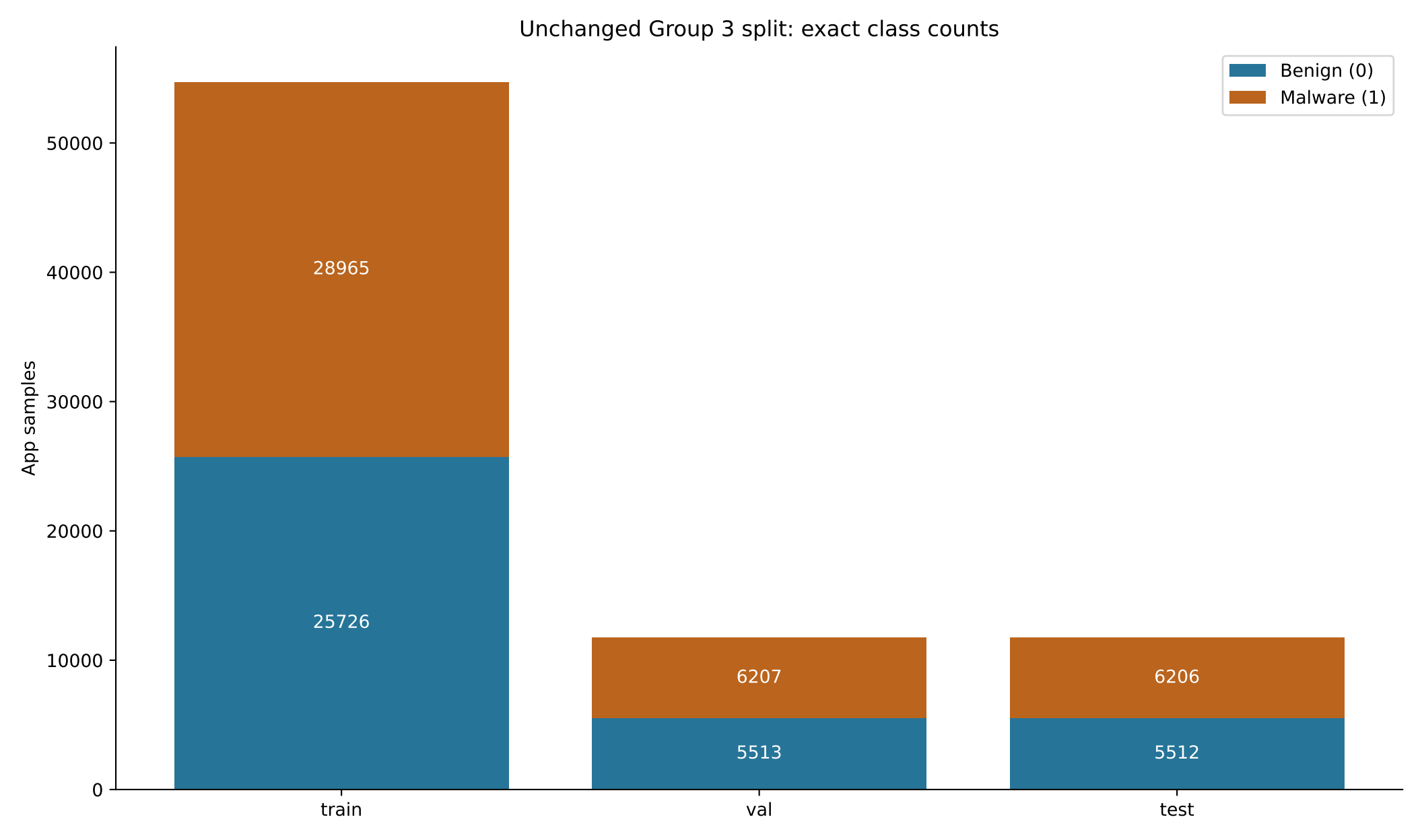

TEST ROC curves: post hoc description only
True positive rate (Malware recall)
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
mi_top50 AUC=0.9596160160563751
mi_top30 AUC=0.9494496699434556
mi_top100 AUC=0.9639141705648404
et_top50 AUC=0.97042641736285
Random ranking reference

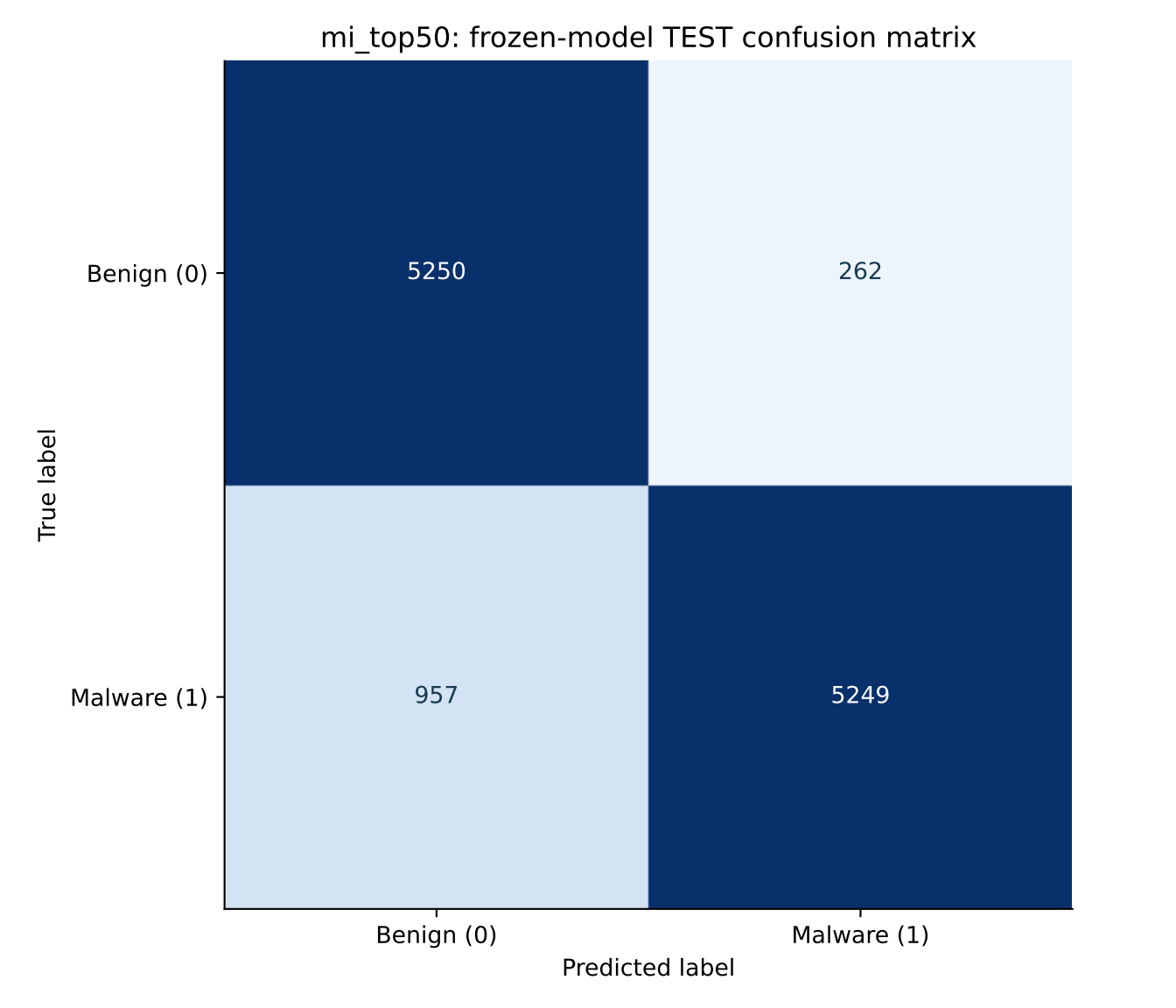

TEST precision-recall curves: post hoc only
Malware precision
Malware recall
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
mi_top50
mi_top30
mi_top100
et_top50
TEST Malware prevalence=0.5296125618706264

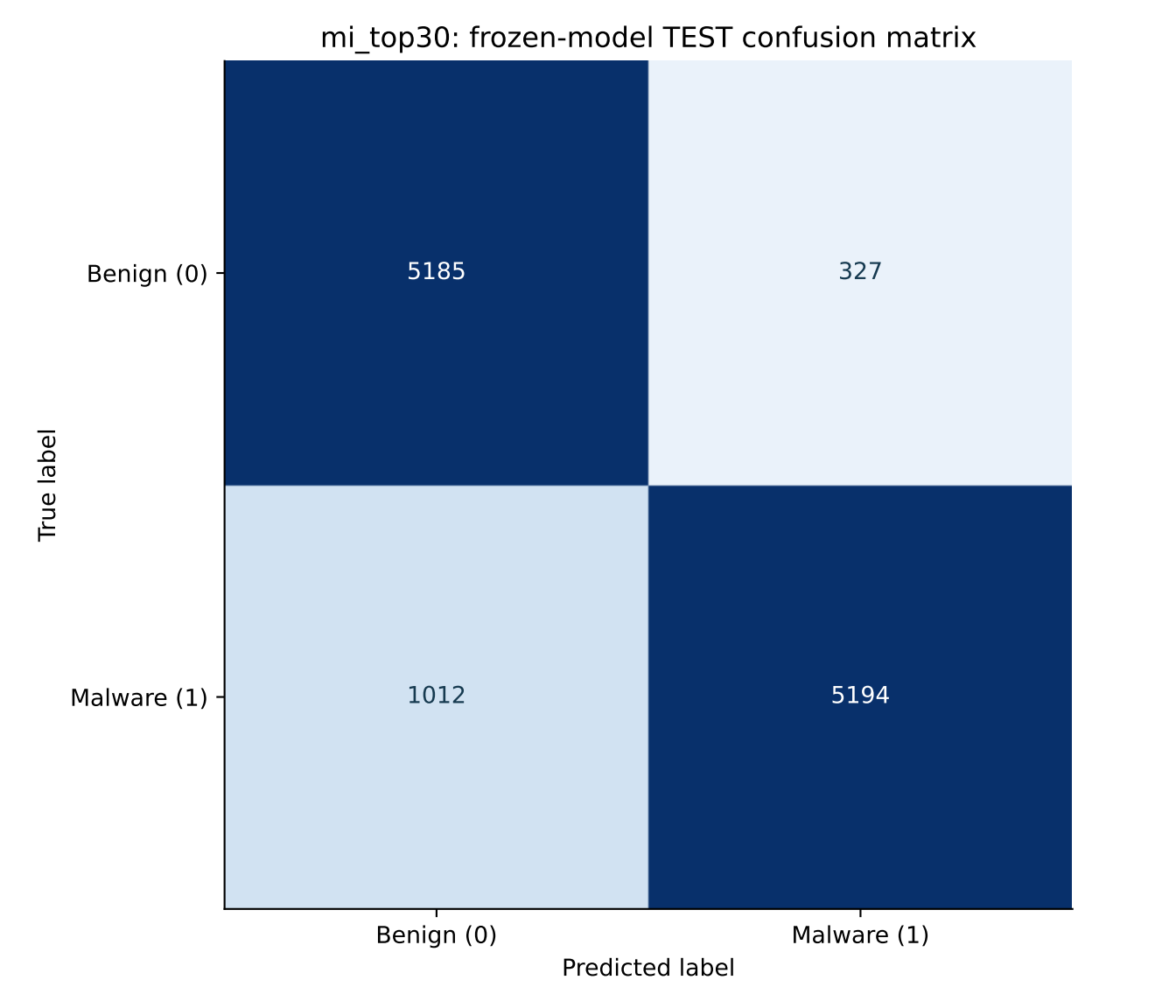

Final model fit only
Measured wall time (seconds)
0.0
0.2
0.4
0.6
0.8
0.15672719999992954
0.06837939999968512
0.5019833999995171
0.13727470000048925
mi_top50
mi_top30
mi_top100
et_top50
Full per-feature-set search
Measured wall time (seconds)
0
1
2
3
4
5
6
7
1.4346088000002055
0.6859557999996468
4.320447699999315
1.361035500000071
mi_top50
mi_top30
mi_top100
et_top50

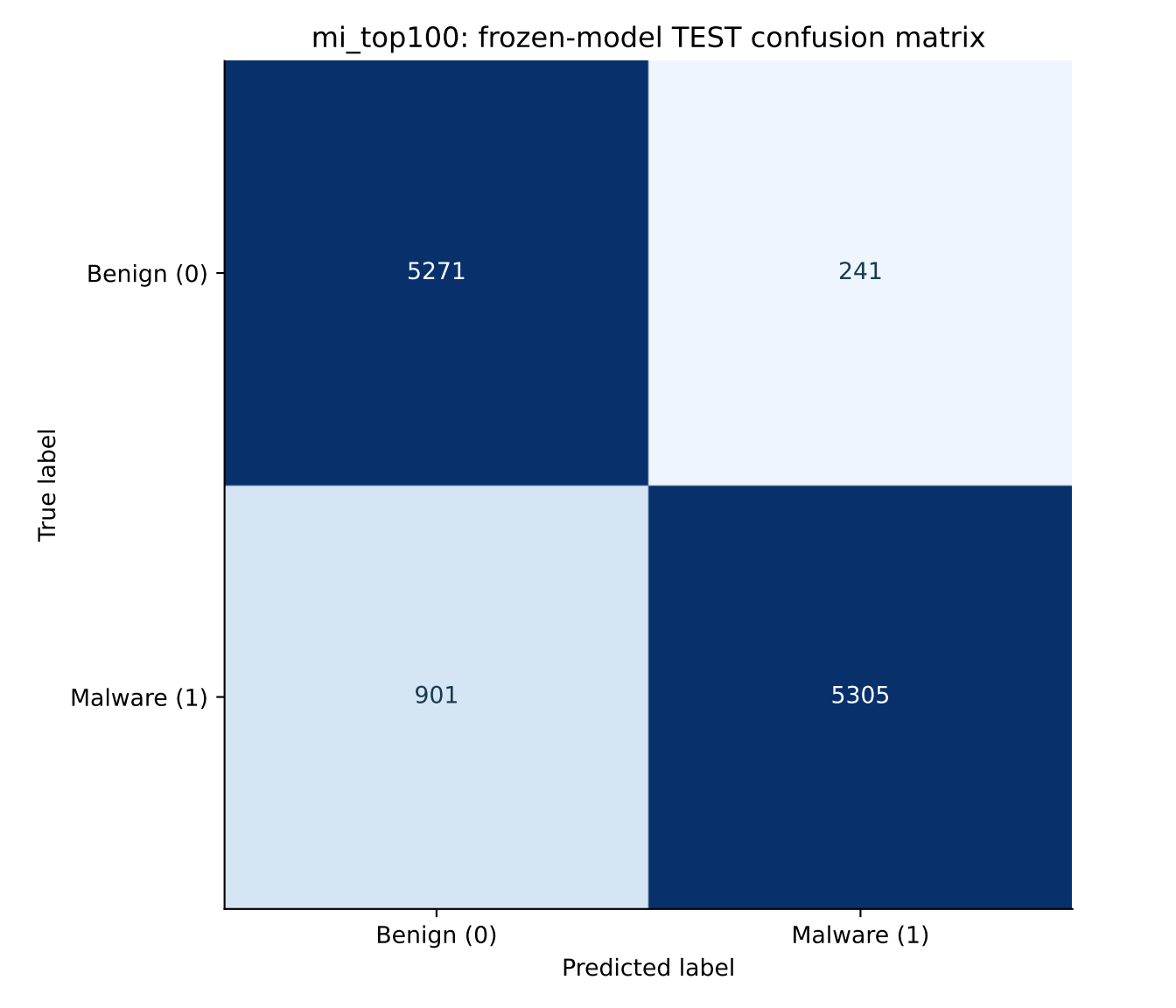

Captured sklearn warnings per fitted candidate
Feature set
mi_top50
mi_top30
mi_top100
et_top50
2
0
reg_param
0.0
0.01
0.05
0.1
0.25
0.5
0.75
1.0
Warning occurrences
2
1
0

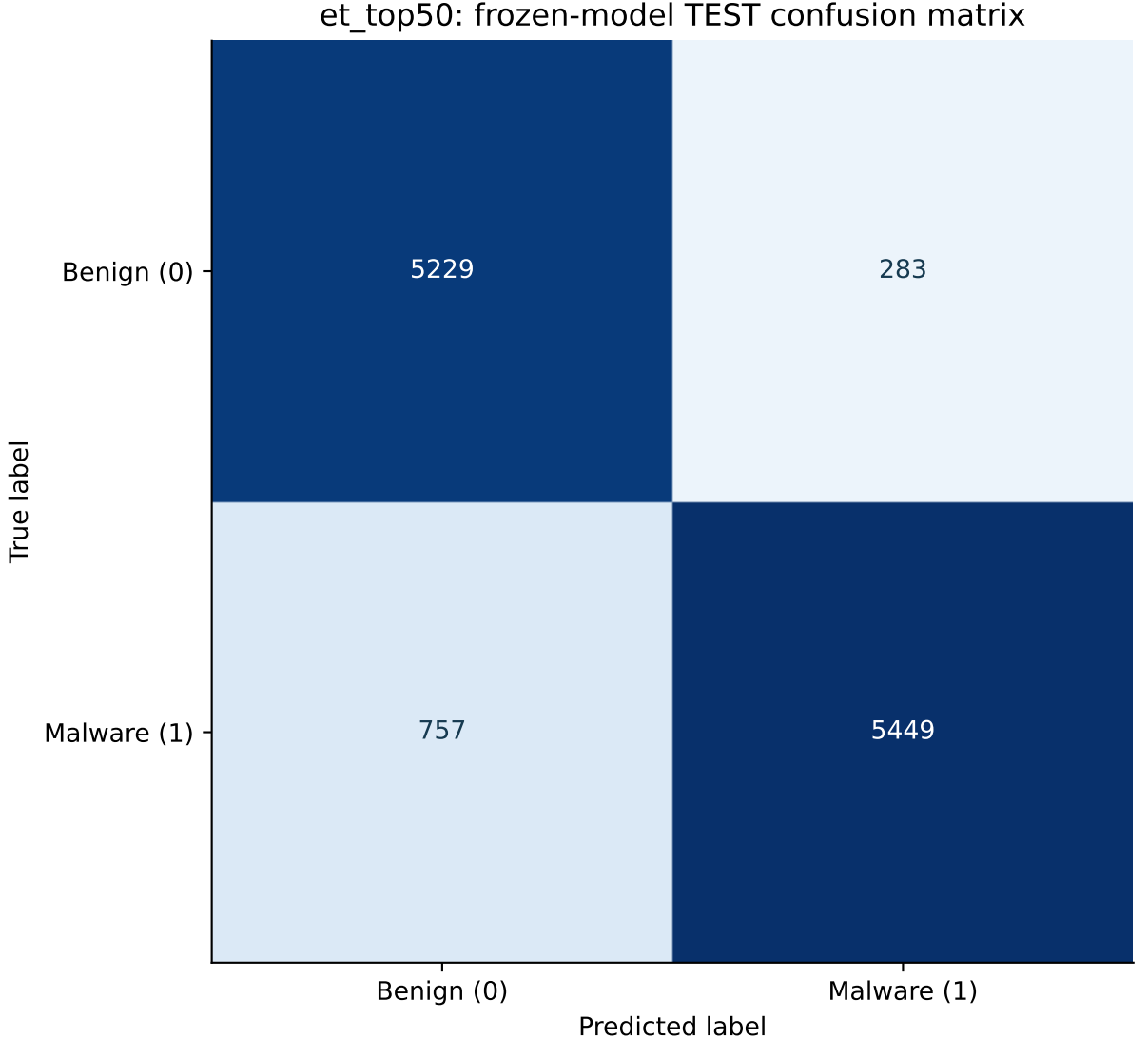

Measured comparison: QDA below GROUP 3 LightGBM
1.0
0.8
0.6
0.4
0.2
0.0
TEST accuracy
TEST Malware F1 (not macro F1)
QDA primary
mi_top50
QDA VAL-selected
et_top50
GROUP 3 LightGBM
replication
GROUP 3 LightGBM
full_standardized


Unchanged Group 3 split: exact class counts
Benign (0)
Malware (1)
App samples
0
10000
20000
30000
40000
50000
28965
25726
6207
5513
6206
5512
train
val
test


mi_top50: frozen-model TEST confusion matrix
Benign (0)
Malware (1)
True label
5250
262
957
5249
Benign (0)
Malware (1)
Predicted label


mi_top30: frozen-model TEST confusion matrix
Benign (0)
Malware (1)
True label
5185
327
1012
5194
Benign (0)
Malware (1)
Predicted label


mi_top100: frozen-model TEST confusion matrix
Benign (0)
Malware (1)
True label
5271
241
901
5305
Benign (0)
Malware (1)
Predicted label


et_top50: frozen-model TEST confusion matrix
Benign (0)
Malware (1)
True label
5229
283
757
5449
Benign (0)
Malware (1)
Predicted label